

Assignment-5

Program 1:

S.No.	Line No.	Error Description	Error Type	Error Correction
1	6	Missing a semicolon ; at the end of the array initialization statement.	Syntax Error	<code>int marks[5] = {80,90,70,60,50};</code>
2	12	Loop runs 6 times (0 to 5), but array <code>marks</code> only has indices 0 to 4.	Array Bounds Error	<code>for(int i=0; i<5; i++)</code>
3	21	Dividing two integers drops decimal values before saving into <code>average</code> .	Type Mismatch	<code>average = (float)total / 5;</code>
4	23	Printing a floating-point variable (<code>average</code>) using an integer specifier (<code>%d</code>).	Type Mismatch	<code>printf("Average = %f\n", average);</code>
5	34	Array size is 5, making index 5 out of bounds (valid indices are 0 to 4).	Array Bounds Error	Remove the line <code>scores[5] = 60;</code>
6	45	The string "Programming" needs 12 bytes, but the array size is capped at 5.	Array Bounds Error	<code>char subject[12];</code>
7	52	Using a string literal "one" as an array index instead of a valid integer.	Type Mismatch	Change index to an integer like 1.
8	54	Using a floating-point number 2.5 as an array index instead of an integer.	Type Mismatch	Change index to an integer like 2.
9	65	The loop condition <code>i<=5</code> prints an out-of-bounds index 5 from the <code>attendance</code> array.	Array Bounds Error	<code>for(int i=0; i<5; i++)</code>
10	76	<code>scanf</code> needs a pointer to store the input value; the address operator is missing.	Pointer Error	<code>scanf("%d", &arr[i]);</code>
11	93	Assigning a string literal "A" to a single character index instead of a literal 'A'.	Type Mismatch	<code>grade[0] = 'A';</code>
12	104	Using a single <code>=</code> overwrites the array value instead of comparing it.	Logical Error	<code>if(prices[2] == 300)</code>

13	113	Trying to copy an entire array pointer directly to another array variable.	Linker / Semantic	Copy elements using a loop or use memcpy.
14	123	Printing a floating-point value (<code>salary[0]</code>) using an integer specifier (<code>%d</code>).	Type Mismatch	<code>printf("%f\n", salary[0]);</code>
15	131	Direct string assignment to a declared character array is illegal in C.	Syntax Error	<code>strcpy(city, "Jaipur");</code>
16	139	The assignment loop condition <code>i<=5</code> writes to an out-of-bounds index.	Array Bounds	<code>for(int i=0; i<5; i++)</code>

Program 2:

S.No.	Line No.	Error Description	Error Type	Error Correction
1	19	Loop reads index 10, outside 0-9 boundaries of <code>numbers[10]</code> .	Array Bounds Error	<code>for(int i=0; i<10; i++)</code>
2	30	Overwrites <code>sum</code> every iteration instead of calculating total.	Logical Error	<code>sum += numbers[i];</code>
3	39	Integer division drops decimals before saving into <code>average</code> .	Type Mismatch	<code>average = (float)sum / 10;</code>
4	41	Uses <code>%d</code> to print a floating-point variable <code>average</code> .	Type Mismatch	<code>printf("Average = %f\n", average);</code>
5	55	Condition checks for lesser values, yielding lowest mark.	Logical Error	<code>if(marks[i] > highest)</code>
6	73	Attempts to write to index 5 on an array of size 5 (0-4).	Array Bounds Error	Remove line <code>attendance[5] = 1;</code>
7	77	Loop execution accesses index 5, causing out-of-bounds error.	Array Bounds Error	<code>for(int i=0; i<5; i++)</code>
8	86	String "ComputerScience" (16 bytes) exceeds array size of 10.	Array Bounds Error	Change to char <code>subject[16];</code>
9	96	Performs direct string literal assignment to character array.	Syntax Error	<code>strcpy(branch, "IT");</code>

10	111	Prints string via %s without a null-terminator \0.	Semantic Error	Add college[5] = '\0';
11	117	Uses string literal "two" instead of valid numeric index.	Type Mismatch	Change index to integer or delete.
12	119	Uses floating-point constant 3.5 instead of integer index.	Type Mismatch	Change index to integer or delete.
13	143	Passes raw element values instead of pointers to scanf.	Pointer Error	scanf("%d", &result[i]);
14	165	Employs assignment = instead of comparison ==.	Logical Error	if(stock[2] == 300)
15	174	Attempts direct baseline assignment of static memory arrays.	Semantic Error	Copy elements individually in a loop.
16	184	Uses %d to output a floating-point array value.	Type Mismatch	printf("%f\n", salary[0]);
17	192	Dynamic assignment loop condition goes to index 5.	Array Bounds Error	for(int i=0; i<5; i++)
18	208	String "JaipurCity" (11 bytes) exceeds storage of 8.	Array Bounds Error	Change to char city[11];
19	218	Direct baseline allocation of one static array to another.	Semantic Error	Copy elements individually in a loop.
20	226	Omit mandatory statement-terminating semicolon.	Syntax Error	printf("Program Ended\n");

Program 3:

S.No.	Line	Error Description	Error Type	Error Correction
1	43	Condition checks for greater values (>), yielding highest mark instead of lowest.	Logical Error	if(marks[i] < lowest)
2	64	Loop checks condition i<=5, accessing out-of-bounds index 5.	Array Bounds Error	for(int i=0; i<5; i++)
3	75	String "Programming" needs 12 bytes including \0, exceeding array boundary of 8.	Array Bounds Error	Change to char subject[12];
4	85	Passing a pointer to an array (&branch) into scanf for %s.	Pointer Error	scanf("%s", branch);

5	102	Explicit assignment to index 5 on a static Array array allocated for 0-4.	Bounds Error	Remove line <code>fees[5] = 100000;</code>
6	116	Uses single assignment <code>=</code> instead of comparison <code>==</code> .	Logical Error	<code>if(stock[2] == 30)</code>
7	125	Attempts direct reference allocation of entire array pointer.	Semantic Error	Copy elements individually inside a loop.
8	135	Uses decimal specifier <code>%d</code> to display floating-point value.	Type Mismatch	<code>printf("%f\n", salary[0]);</code>
9	145	Missing address reference operator <code>&</code> within <code>scanf</code> .	Pointer Error	<code>scanf("%d", &result[i]);</code>
10	161	Direct string assignment to a previously allocated array.	Syntax Error	<code>strcpy(city, "Jaipur");</code>
11	171	Uses a string literal "one" as an array index.	Type Mismatch	Change index to an integer.
12	173	Uses a floating-point value 2.5 as an array index.	Type Mismatch	Change index to an integer.
13	185	Print statement loop condition evaluates index 5.	Array Bounds Error	<code>for(int i=0; i<5; i++)</code>
14	199	Printing <code>%s</code> without a terminal null byte character <code>\0</code> .	Semantic Error	Add <code>college[5] = '\0';</code>
15	205	Attempts raw base reference instance assignment of an array.	Semantic Error	Copy elements individually inside a loop.
16	215	String "Computer" (9 bytes) overflows boundary size of 6.	Array Bounds Error	Change to <code>char course[9];</code>
17	225	Loop tracks past upper array boundaries (index 5).	Array Bounds Error	<code>for(int i=0; i<5; i++)</code>
18	240	Omitted statement terminating semicolon.	Syntax Error	<code>return 0;</code>

Program 4:

S.No.	Line No.	Error Description	Error Type	Error Correction
1	23	Printing a floating-point variable (averageSales) using an integer specifier (<code>%d</code>).	Type Mismatch	<code>printf("Average Sales = %f\n", averageSales);</code>

2	31	Dividing two integers drops the precision values before storing them in a float.	Type Mismatch	<code>averageSales = (float)totalSales / 5;</code>
3	35	Condition checks for lesser values (<), yielding the lowest sale instead of highest.	Logical Error	<code>if(sales[i] > highestSale)</code>
4	55	Explicitly assigns to index 5 on a static array allocated with a capacity of 5 (0-4).	Array Bounds Error	Remove the line <code>monthlySales[5] = 250;</code>
5	59	The printing loop condition <code>i <= 5</code> reads out-of-bounds index 5 from <code>monthlySales</code> .	Array Bounds Error	<code>for(int i=0; i<5; i++)</code>
6	68	String "LaptopComputer" requires 15 bytes including <code>\0</code> , overflowing array size of 8.	Array Bounds Error	Change declaration to <code>char product[15];</code>
7	76	Performs direct string literal pointer assignment to a previously allocated array variable block.	Syntax Error	<code>strcpy(category, "Electronics");</code>
8	84	Uses a string literal "first" as an indexing position marker inside the brackets.	Type Mismatch	Delete line or change index to an integer.
9	86	Uses a floating-point decimal constant <code>2.7</code> as an array indexing coordinate.	Type Mismatch	Delete line or change index to an integer.
10	106	Missing address reference operator <code>&</code> within the string sequence parsing execution.	Pointer Error	<code>scanf("%d", &orders[i]);</code>
11	126	Employs assignment operator <code>=</code> instead of comparison equality operator <code>==</code> .	Logical Error	<code>if(revenue[2] == 3000)</code>
12	135	Attempts direct reference allocation of one static memory array block to another.	Semantic Error	Copy data elements individually inside a loop.
13	145	Uses integer format specifier <code>%d</code> to display a floating-point value (<code>commission</code>).	Type Mismatch	<code>printf("%f\n", commission[0]);</code>
14	153	Performs direct initialization string assignment to a declared character array.	Syntax Error	<code>strcpy(city, "Jaipur");</code>
15	166	Prints character collection via <code>%s</code> without manually allocating a trailing null-terminator <code>\0</code> .	Semantic Error	Add <code>branch[4] = '\0';</code> before printing.

16	173	Modification loop tracks past upper array boundaries by executing through index 5.	Array Bounds Error	<code>for(int i=0; i<5; i++)</code>
17	188	String "Marketing" requires 10 bytes including <code>\0</code> , exceeding the array size of 6.	Array Bounds Error	Change declaration to <code>char department[10];</code>
18	196	Attempts raw base reference instance assignment of array <code>sales</code> to array <code>expenses</code> .	Semantic Error	Copy data elements individually inside a loop.
19	204	Uses a negative integer value <code>-1</code> as an index, causing an out-of-bounds stack corruption.	Array Bounds Error	Delete or fix index coordinate.
20	222	Uses string specifier <code>%s</code> to output an integer value (<code>feedback[0]</code>).	Type Mismatch	<code>printf("%d\n", feedback[0]);</code>
21	226	Omitted statement terminating semicolon.	Syntax Error	<code>printf("Report Generated Successfully\n");</code>

Program 5:

S.No.	Line No.	Error Description	Error Type	Error Correction
1	23	Printing a floating-point variable (<code>averageSales</code>) using an integer specifier (<code>%d</code>).	Type Mismatch	<code>printf("Average Sales = %f\n", averageSales);</code>
2	31	Dividing two integers drops the precision values before storing them in a float.	Type Mismatch	<code>averageSales = (float)totalSales / 5;</code>
3	35	Condition checks for lesser values (<code><</code>), yielding the lowest sale instead of highest.	Logical Error	<code>if(sales[i] > highestSale)</code>
4	55	Explicitly assigns to index 5 on a static array allocated with a capacity of 5 (0-4).	Array Bounds Error	Remove the line <code>monthlySales[5] = 250;</code>
5	59	The printing loop condition <code>i <= 5</code> reads out-of-bounds index 5 from <code>monthlySales</code> .	Array Bounds Error	<code>for(int i=0; i<5; i++)</code>

6	68	String "LaptopComputer" requires 15 bytes including \0, overflowing array size of 8.	Array Bounds Error	Change declaration to <code>char product[15];</code>
7	76	Performs direct string literal pointer assignment to a previously allocated array variable block.	Syntax Error	<code>strcpy(category, "Electronics");</code>
8	84	Uses a string literal "first" as an indexing position marker inside the brackets.	Type Mismatch	Delete line or change index to an integer.
9	86	Uses a floating-point decimal constant 2.7 as an array indexing coordinate.	Type Mismatch	Delete line or change index to an integer.
10	106	Missing address reference operator & within the string sequence parsing execution.	Pointer Error	<code>scanf("%d", &orders[i]);</code>
11	126	Employs assignment operator = instead of comparison equality operator ==.	Logical Error	<code>if(revenue[2] == 3000)</code>
12	135	Attempts direct reference allocation of one static memory array block to another.	Semantic Error	Copy data elements individually inside a loop.
13	145	Uses integer format specifier %d to display a floating-point value (commission).	Type Mismatch	<code>printf("%f\n", commission[0]);</code>
14	153	Performs direct initialization string assignment to a declared character array.	Syntax Error	<code>strcpy(city, "Jaipur");</code>
15	166	Prints character collection via %s without manually allocating a trailing null-terminator \0.	Semantic Error	Add <code>branch[4] = '\0';</code> before printing.
16	173	Modification loop tracks past upper array boundaries by executing through index 5.	Array Bounds Error	<code>for(int i=0; i<5; i++)</code>
17	188	String "Marketing" requires 10 bytes including \0, exceeding the array size of 6.	Array Bounds Error	Change declaration to <code>char department[10];</code>
18	196	Attempts raw base reference instance assignment of array <code>sales</code> to array <code>expenses</code> .	Semantic Error	Copy data elements individually inside a loop.
19	204	Uses a negative integer value -1 as an index, causing an out-of-bounds stack corruption.	Array Bounds Error	Delete or fix index coordinate.

20	222	Uses string specifier %s to output an integer value (feedback[0]).	Type Mismatch	printf("%d\n", feedback[0]);
21	226	Omitted statement terminating semicolon.	Syntax Error	printf("Report Generated Successfully\n");